SALTDRAG · MARKET AND VIABILITY

SaltDrag market and viability

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

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0.1	2026-10-24	Amish Chadha	First market and viability note (kit 1.8 upgrade); prices checked 2026-10-25

SaltDrag market and viability

SaltDrag is for families who make salt by hand in small pans, first the Agariyas of the Little Rann of Kutch, who may not use machines and earn a small share of the price of their salt; nothing on the market gathers salt from the bund by hand, and the nearest bought parts (a hand winch, a rope puller) are built for lifting or occasional pulling, not for a season in brine. An open design lets producer groups and town workshops build, repair and adapt it without a supplier. Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Inland hand-made salt	Agariya families in the Little Rann of Kutch, India; about 60,000 Agariyas produce 30 per cent of India's inland salt (Mongabay India, 2021)	The dantala hand scraper they use today (price to assess); no hand-powered gathering tool on the market	2,897 a set (bom/bom.csv), one set per family pan	Gather from the dry bund instead of standing in the brine; within the access conditions because it is hand powered
Producer groups and NGOs	Women-led producer organisations and livelihood programmes in Kutch, starting with SEWA (candidate, nothing agreed)	Solar brine pumps, the one recent hardware change that spread: about 80 per cent of 7,000 Agariya families have at least one (Down To Earth)	2,897 a set for a pilot; cost per family equals cost per set	A costed, repairable kit with a safety case they can fund and train on
Bought parts as an alternative	Families or groups who would buy a winch rather than build a capstan	Hand winches in India: 1 t galvanised worm gear winch INR 3,900; 800 kg hand winch INR 2,480 (IndiaMART, checked 2026-10-25). Fibre-rope puller, 3/4 t: USD 82.23 (Knot and Rope Supply, checked 2026-10-25)	Not comparable: these are lifting winches with a short wire rope or occasional-use pullers; none has a pawl, brake and drum sized for 45 m of 12 mm rope and a season in brine	SaltDrag is a complete system (capstan, roller, rake, rope, anchors) sized and checked for this job
Coastal and artisanal salt	Small coastal pans in Kutch creeks and elsewhere; artisanal and specialty salt makers	Industrial salt harvesters (price to assess; built for large mechanised works)	2,897 a set	Gentle gathering of large crystals from shallow pans without a machine

Main cost lines of option A, from the bill of materials: fabrication labour and galvanising USD 450, side frames USD 220, drums USD 244, gears USD 190, rake blade USD 190, pull ropes USD 190. For comparison, 100 m of 12 mm double braid polyester rope (breaking strength about 2,957 kg) costs GBP 140 from a UK rope maker (RopeServices UK, checked 2026-10-25), in line with the USD 95 allowed for each 45 m rope. The lower-cost concept options are about USD 1,400 (option B) and USD 1,800 (option C) a set (SDG-OPT-001).

Open route

SaltDrag spreads as a design, not a product. A producer group or NGO commissions a town workshop near the Rann (Gandhidham, Dhrangadhra or Kharaghoda) to build a set from the drawings, the making sketches and the build plan; gears, bearings, anchors and rope are bought from stockists. The partner trains crews and keeps spares. Everything is published under CERN-OHL-S-2.0, so any improvement made by a workshop or a group stays open, and the hand capstan is offered as a common block with SiltHaul so that one proven capstan serves both. No sale, margin or certification is planned at TRL 3: SaltDrag is published as an open engineering reference, not certified equipment.

Sources

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- Knot and Rope Supply, Maasdam rope puller (rope ratchet), 3/4 ton, USD 82.23, checked 2026-10-25. https://knotandrope.com/products/maasdam-rope-puller-rope-ratchet
- RopeServices UK, 12 mm black double braid polyester rope, 100 m reel, GBP 140.00 (one to four reels), checked 2026-10-25. https://www.ropeservicesuk.com/product/12mm-black-double-braid-polyester-rope-100-metre-reel/